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Problem Statement

Question : Sumit is 3 times as old as his son. Five years later, he shall be two and a half times as old as his son. How old is Sumit at present ?

Intersection of lines and Age

Solution:

Let Age of son = x

Age of Father = $y = 3x$

$$3x - y = 0 \quad (3.1)$$

Five years later, father shall be two and a half times as old as his son, So

$$y + 5 = \frac{5}{2}(x + 5) \quad (3.2)$$

$$5x - 2y = -15 \quad (3.3)$$

The system of equations formed is :

$$3x - y = 0 \quad (3.4)$$

$$5x - 2y = -15 \quad (3.5)$$

Writing it in the matrix form,

$$\begin{pmatrix} 3 & -1 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ -15 \end{pmatrix} \quad (3.6)$$

Forming the augmented matrix to solve the system of equations,

$$\left(\begin{array}{cc|c} 3 & -1 & 0 \\ 5 & -2 & -15 \end{array} \right) \quad (3.7)$$

Using Gaussian Elimination,

$$\left(\begin{array}{cc|c} 3 & -1 & 0 \\ 5 & -2 & -15 \end{array} \right) \xleftrightarrow{R_2 \rightarrow 3R_2 - 5R_1} \left(\begin{array}{cc|c} 3 & -1 & 0 \\ 0 & -1 & -45 \end{array} \right) \quad (3.8)$$

Using back substitution we get,

$$-y = -45 \quad (3.9)$$

$$y = 45 \quad (3.10)$$

$$3x - y = 0 \quad (3.11)$$

$$3x = 45 \quad (3.12)$$

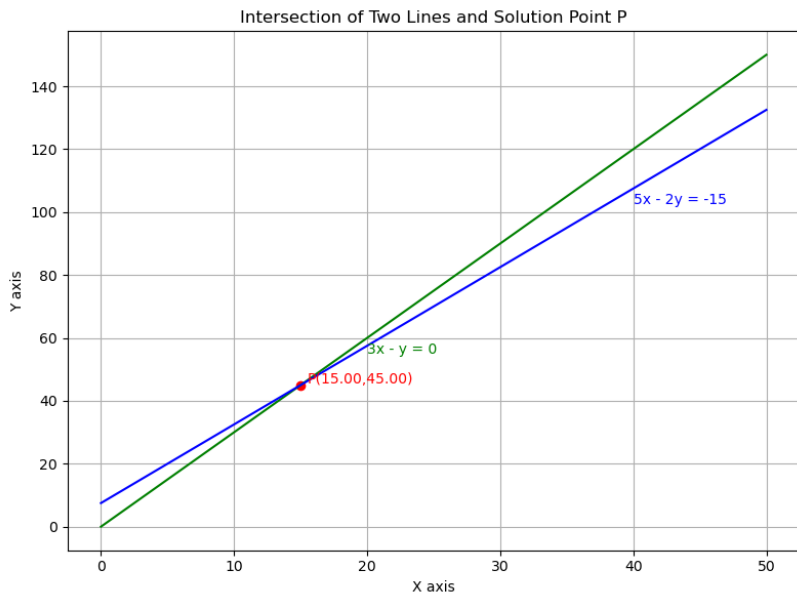
$$x = 15 \quad (3.13)$$

The solution for the system of equations is :

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 15 \\ 45 \end{pmatrix} \quad (3.14)$$

Therefore the present age of father = $y = 45$

Plots



C Code

```
#include <stdio.h>

int main() {
    float son, sumit;

    // Problem:
    // Sumit is 3 times as old as his son
    // => sumit = 3 * son
    //
    // After 5 years:
    // sumit + 5 = 2.5 * (son + 5)

    // Substituting sumit = 3 * son:
    // 3*son + 5 = 2.5*(son + 5)
    // 3*son + 5 = 2.5*son + 12.5
    // 0.5*son = 7.5
    // son = 15
    // sumit = 45
```

```
son = (12.5f - 5.0f) / (3.0f - 2.5f); // solving step
sumit = 3 * son;

printf("Son's present age = %.0f years\n", son);
printf("Sumit's present age = %.0f years\n", sumit);

return 0;
}
```

Python Code 1: call_c.py

```
import ctypes

# Load the shared object file
lib = ctypes.CDLL("./sumit_age.so")

# Define return types for functions
lib.son_age.restype = ctypes.c_int
lib.sumit_age.restype = ctypes.c_int

# Call functions from C
son = lib.son_age()
sumit = lib.sumit_age()

# Print results
print("Son's present age =", son, "years")
print("Sumit's present age =", sumit, "years")
```

Python Code 2

```
import os
import numpy as np
import matplotlib.pyplot as plt

# save figure in figs folder
figs_folder = os.path.join("../", "figs")
os.makedirs(figs_folder, exist_ok=True) # create folder if it
    doesn't exist

# solve system of equations directly using numpy
#  $3x - y = 0$ 
#  $5x - 2y = -15$ 
A = np.array([[3, -1],
               [5, -2]], dtype=float)

b = np.array([0, -15], dtype=float)

solution = np.linalg.solve(A, b)
x, y = solution
```

```
print("Solution from Python:", solution)
# create x range for plotting
x_vals = np.linspace(0, 50, 100)

# equations
y1 = 3 * x_vals - 0 # from  $3x - y = 0$   $y = 3x$ 
y2 = (5 * x_vals + 15)/2 # from  $5x - 2y = -15$   $y = (5x + 15)/2$ 

# plotting
plt.figure(figsize=(8, 6))
# plot the lines
plt.plot(x_vals, y1, color="green")
plt.plot(x_vals, y2, color="blue")

# plot the solution point
plt.scatter(x, y, color="red")
plt.text(x + 0.5, y + 0.5, f"P({x:.2f},{y:.2f})", fontsize=10,
        color="red")

# move line labels towards right
x_label1 = 20 # moved right from 10 20
y_label1 = 3 * x_label1 - 5 # slight downward offset for clarity
```

```
plt.text(x_label1, y_label1, "3x - y = 0", color="green",
         fontsize=10, rotation=0)

x_label2 = 40 # moved right from 30 40
y_label2 = (5 * x_label2 + 15)/2 - 5 # slight downward offset
plt.text(x_label2, y_label2, "5x - 2y = -15", color="blue",
         fontsize=10, rotation=0)

# labels and grid
plt.xlabel("X axis")
plt.ylabel("Y axis")
plt.title("Intersection of Two Lines and Solution Point P")
plt.grid(True)

# save figure
plt.tight_layout()
plt.savefig(os.path.join(figs_folder, "solution.png"))
plt.show()
```